AI-Based Location-Adaptive Landslide Early-Warning Using Abnormal Environmental Change and Temporal Risk Escalation

Abstract*— *There are many different factors that affect the occurrence of landslides. These factors include rainfall levels and variables, moisture of the soil, topography, geology, land use as well as the other local environment. As those factors can change over time, the warning system must consider not just one risk indicator but analyze how the risk itself is likely to change over time. The paper suggests an early warning framework that uses artificial intelligence and adapts it to the needs of the users regarding the present risk situation. In particular, all the past observations of the environmental situation are translated into time frames and used for assessment. Furthermore, the current situation is compared to the one observed at the same place previously and examined to see whether the current risk level undergoes steady increases. Based on the obtained information, alerts are issued, and the system registers valuable indicators such as the time of detection, lead time, false alarms, missed events, uncertainty, and consistency of the information. Besides, the what-if analysis is carried out in order to find out the consequences of fluctuations of the environment for landslide risk change.

Index Terms— Landslide, artificial intelligence, machine learning, early warning, temporal risk, abnormal change detection, lead time, uncertainty.

I. Introduction

Landslides can arise very quickly when the right combinations of rainfall and previous wetness, along with aspects like the slope, soil features, drainage systems, and vegetation are present. Such conditions are geographically variable and may change with time. For this reason, a good warning system should do more than just point the areas that are prone to landslide activity in general. It should monitor whether the area is trending in the right direction.

Traditional methods to establish the areas with potential danger zones use susceptibility maps. These maps are helpful to know where there is a risk but cannot provide information on the trends indicating whether the area actually starts experiencing risks.

Nowadays, machine learning methods are actively used for landslides risk prediction since they can model complicated relationships between environmental variables. Using RS (remote sensing) data also helps to evaluate rainfall, vegetation, surface characteristics, and other elements over a big territory. Nevertheless, an efficient early warning system should address very practical questions that are not related to prediction.

In order to resolve these problems, the proposed work consists of four interconnected steps: multi-temporal feature extraction, AI-based risk assessment, location-based anomaly detection, and risk escalation over time. It follows that the system provides not only a risk probability but also a warning status based on the last risk behavior. Put differently, we go away from asking the question "What's the risk at the moment?" to the question "Is the risk increasing in a significant way and justifying a warning?"

The contributions of this work are as follows: (1) a location-based baseline for detection of unusual environmental situations; (2) the temporal risk escalation layer enabling transformation of a series of predictions into comprehensible warnings; (3) event-oriented evaluation accounting for lead time, false alarms, and missed events besides ordinary evaluation metrics; (4) the experimental approach including ablation, uncertainty, and stability as well as what-if scenarios.

The contribution of this study is presented as a framework-level integration and assessment approach instead of stating that each component itself is novel.

II. Related Work

A. Landslide Susceptibility and Prediction

Research on slope instability often involves different aspects like topography, geology, rainfall, land usage, among others. Both statistical techniques and models based on machine learning have demonstrated the potential of nonlinear relations between these variables and actual landslide occurrences for predicting slope instability [1], [3]. Recent literature reviews indicate growing popularity of advanced methods, such as ensemble techniques and deep learning, but they still face difficulties regarding data quality, their applicability to different locations, and validation [6], [7].

B. Rainfall and Soil-Moisture Based Warning

Rainfall thresholds have been utilized as landslide alert systems for many years considering that heavy and prolonged rainfall increases pore-water pressure and decreases slope stability [2]. Soil moisture provides additional information about ground wetness prior to the measurement. Different satellite products like IMERG are used to measure rainfall over vast areas while remote-sensing data is useful in assessing environmental conditions for a longer period of time [5].

C. Research Gap

The gap in research that is discussed here does not refer to an absence of models for predicting landslides. Rather, there is a necessity for a system that can be replicated in order to connect the ecological condition of the present time with the predicted risk recurrence and the value of the warning. The system presented here integrates all these aspects together while leaving open the possibility of modifying the prediction methodology.

III. Problem Formulation

Take one location l at time t. At this time, the variable $X_{l,t}$ represents observations of the environment at this place and $Y_{l,t}$ is the event label. Therefore, the prediction layer computes

$$P_{l,t} = f(X_{l,t-L:t}). \tag{1}$$

Here, L is the size of the look-back period and $P_{l,t}$ stands for the chance of landslide occurrence within the required period of forecasting. However, instead of treating every point as an independent variable, the suggested method creates a chain out of the latest predictions $P_{l,t-k:t}$ and monitors how the risk changes over time.

For each situation, a historical baseline B_l is created using appropriate period without events. Thus, the extent of deviation from this baseline can be measured with the help of abnormality score:

$$A_{l,t} = D(X_{l,t}, B_l). \tag{2}$$

where $D(\cdot)$ refers to a standardized measure of discrepancy. Alert only gets issued when both predicted risk and the temporal evidence meet the conditions established during validation. This enables the avoidance of giving an alert in response to one erroneous prediction.

IV. Proposed Framework

Fig. 1 summarizes the proposed workflow.

A. Data Acquisition

The framework is capable of using rainfall, soil moisture content, temperature, topography, land-use patterns and other factors that can be collected in the study area. Information gathered from landslide inventory corresponds to the dates and locations of landslide occurrences useful for supervised learning and event-based assessments. The final choice of data should be based on the geographical features of the study area, time resolution required and availability of the data.

B. Preprocessing and Feature Construction

Missing observations are dealt with based on the temporal behavior of each feature. Continuous features are standardized based on the statistics that are only computed using the training dataset, while the categorical features are stored consistently throughout the entire dataset. Furthermore, it is imperative that any information referring to the future does not appear in the features corresponding to any previous times of prediction.

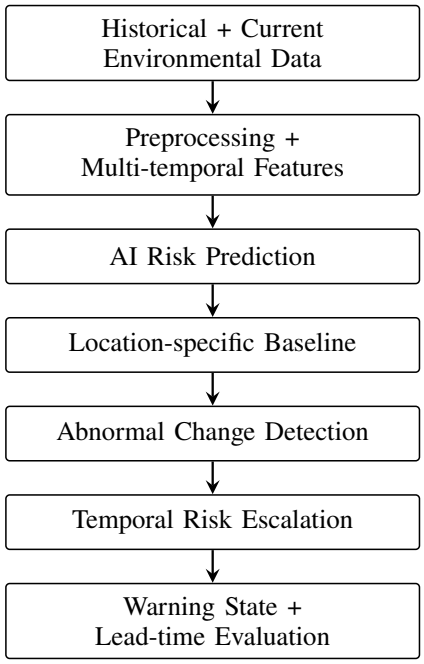


Fig. 1. Overall workflow of the proposed location-adaptive temporal early-warning framework.

TABLE I
CORE DATA VARIABLES AND THEIR ROLES

Variable	**Example Representation**	**Role**
Rainfall	Hourly/daily accumulation	Trigger intensity
Antecedent rainfall	Rolling 3–30 day totals	Prior wetness
Soil moisture	Normalized moisture index	Saturation state
Slope	Degrees or percentage	Terrain condition
Elevation	Digital elevation model	Topographic context
Land cover	Categorical classes	Surface condition
Event record	Date and location	Target/evaluation

The feature set includes instantaneous observations, rolling measures, moving averages, deviations, and short-term trends. For example, rainfall would be characterized by the amount of rain falling now and the amount of rain that fell before. Similarly, the soil moisture would be identified through the current level of moisture as well as the changes that have occurred in the past.

C. AI Risk Prediction

The prediction layer may consist of a supervised classification method or a probabilistic ensemble. As far as the student-level prototype is concerned, decision-tree-type methods are suitable since they are capable of being used for different types of

predictors and also provide essential information regarding the contribution of the predictors to the model. The model is trained on time-separated datasets and generates $P_{l,t}$ based on the selected forecasting horizon.

D. Location-Adaptive Abnormal Change Detection

Using a single threshold for all locations results in conflicting alarms. A certain level of rainfall, or soil moisture, which is regarded as "normal" in one location, and "abnormal" in another. To eliminate such inconsistency, the framework operates with separate threshold for each location. The computation of abnormality score is based on the comparison of current conditions with the past performance of a given location. Depending on the quality of the data used, a particular location can be characterized by median and interquartile range or mean and standard deviation.

E. Temporal Risk Escalation

The recent risk values are analyzed in terms of trajectory instead of as independent forecasts. One can implement a simpler trend approach over a rolling window as shown below:

$$\hat{\beta}_{l,t} = \frac{P_{l,t} - P_{l,t-w+1}}{w-1}. \tag{3}$$

A warning is not raised simply when $P_{l,t}$ crosses a certain threshold level. The system also checks if the risk value is high, if the observed trend is growing, and if the upward movement persists for some defined number of observations. The need for persistence based on the validation process reduces the sensitivity of the warning logic.

Algorithm 1: Temporal Risk Escalation
Input: Risk sequence P_t, window w, risk threshold τ_p, trend threshold τ_β, persistence q.
Output: Warning state W_t.
1. Get the recent risk forecast P_t.
2. Refresh the rolling risk buffer.
3. In case there exist less than w observations, return to NORMAL.
4. Estimate recent trend β_t.
5. Calculate location adaptive abnormality score.
6. Count consecutive increasing observations.
7. If conditions of risk, trend, abnormality and persistence meet, return ALERT.
8. Else if intermediate condition is fulfilled, return WATCH.
9. In any other case return to NORMAL.
10. Store the state for event based evaluation.

F. Warning States and Lead Time

To facilitate interpretation, the system makes use of three warning states which are defined as NORMAL, WATCH and ALERT. The transition rules are determined by means of training and validation datasets but not by the final test events. If an event happens at T_e and its associated first valid warning is issued at T_w, the available lead time is given as:

TABLE II
Experimental Design

Component	Design
Training	Earlier historical period
Validation	Later period for threshold/model selection
Testing	Unseen final period
Temporal window	Multiple recent observations
Spatial test	Optional unseen locations
Target	Event within defined forecast horizon

$$LT = T_e - T_w. \tag{4}$$

This gives a useful indication of how much time was available by the system between its first sustained warning and the occurrence of an event.

V. Experimental Design

A. Dataset Splitting

When the data has a temporal structure, the data is divided into segments in the order of time rather than through randomization. The training segment comes before the validation part and the testing part. If there are observations from multiple geographic locations, a geographical test segment may be used to check whether these patterns can be useful in locations where the model was not trained earlier.

B. Evaluation Metrics

The results can be measured using a number of metrics including precision, recall, F1-score, ROC-AUC, PR-AUC, and calibration criteria. However, since we are dealing with the early warning problem, we should add such metrics as event detection rate, median lead time, number of false alarms, and missed events.

C. Ablation Study

The role of each layer is analyzed through the comparison of four configurations: only prediction, prediction and temporal escalation, prediction and abnormal-change detection, and the full framework. The comparison shows us where the improvement comes from instead of crediting the performance to an unknown combination of processing steps.

D. Uncertainty and Sensitivity

Uncertainty can be analyzed using ensemble variation, bootstrap sampling, probability calibration, or prediction intervals if applicable. In addition, sensitivity tests involve modifying the set of input variables or creating controlled missing values to measure the effect on the warning behavior. These studies show whether the warning associated with the system is stable or it depends heavily on some particular variable.

TABLE III
EVALUATION METRICS

Metric	Purpose	Desired Interpretation
Precision	Correct warnings among warnings	Higher is better
Recall	Detected events among events	Higher is better
F1-score	Balance of precision and recall	Higher is better
PR-AUC	Imbalanced-event discrimination	Higher is better
Lead time	Warning advance time	Larger, useful lead time
False alarms	Warnings without events	Lower is better
Missed events	Events without valid warning	Lower is better
Calibration	Reliability of probabilities	Better alignment

TABLE IV
ABLATION CONFIGURATIONS

ID	AI Prediction	Temporal/Change Layer	Purpose
A	Yes	No	Baseline
B	Yes	Temporal only	Test persistence
C	Yes	Abnormal change only	Test local baseline
D	Yes	Both	Complete framework

VI. System Implementation

The prototype is structured in the form of numerous elementary modules. The data module contains time-stamped data arrays and event labels while the pre-processing module provides quality feature arrays. The model module generates risk probabilities. The warning engine monitors up-to-date predictions and warning states. The dashboard enables the recent information outputs to be seen together, including the location map, current risk, risk evolution, indications of abnormalities, and historical information about warnings.

VII. Model Development and Validation

A. Candidate Models

To ensure fair comparison between the various models, the prediction layer must be kept separate from all of them. The prediction layer permits logistic regression as a basic and clear model, whereas random forest and gradient boosted trees are used for their ability to recognize complex relations between rainfall, soil moisture, terrain, and independent time features.

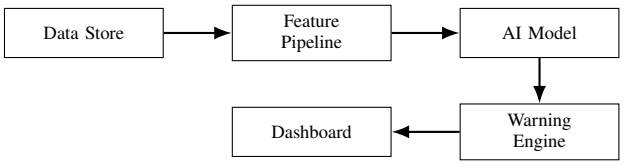


Fig. 2. Modular architecture used for the prototype implementation.

In case of sufficiently large datasets, one may use a recurrent or any other temporal neural model as well. It is important to mention that the change of prediction model does not imply any adjustments in the warning logic.

B. Hyperparameter Selection

All hyperparameters of the model are determined with the help of training and validation data only. Tuning processes, e.g. grid search or randomized search, may include estimating tree depth, number of estimators, learning rate, or minimum samples requirements. The test data is kept separate from the process until the final evaluation is carried out. The importance of separation comes from the fact that if the parameters or warnings are chosen based on test events the reported results will look better than reality.

C. Probability Calibration

A machine learning score does not necessarily imply that it is a well-calibrated probability. Reliability diagrams and measures like the Brier score can be used to assess calibration. If calibration is required, techniques such as Platt scaling or isotonic regression can be applied to the validation set. The obtained probability will then be forwarded to the time alert system. This makes clear what it means to say ”the probability of a risk is equal to 0.70”, but at the same time, one can understand that the model probability is not an equivalent of certainty.

D. Class Imbalance

Landslides occur much less frequently compared to the instances of their absence. Therefore, if this imbalance is not accounted for, a classifier will most likely be biased towards the predominating class. Thus, comparisons of techniques like class weighting, balanced sampling, and threshold adjustment can be made in the research. Precision-recall analysis is especially effective in situations where the class of interest is rare. Any type of resampling must be done only on the training dataset to exclude information leakages in the process of testing and validation.

VIII. Temporal Risk Representation

A. Rolling Features

The system generates features at every point in time using data from multiple recent intervals. The use of short intervals provides the opportunity to observe rapid changes, while longer intervals provide the opportunity to study cumulative load on the environment. Rainfall rates during a 1-day, 3-day, 7-day, and 30-day period, soil moisture changes, and temperature and vegetation moving average values are examples of features

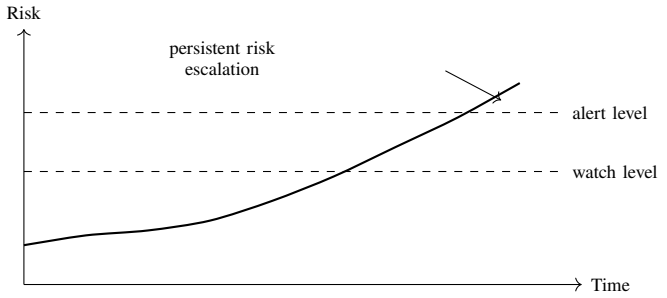


Fig. 3. Illustration of the risk trajectory used by the temporal escalation layer.

generated. The final intervals need to match the temporal resolution of the dataset and be validated through tests.

B. Risk Trajectory

The main result of the system that is being proposed will be in the form of a risk trajectory rather than just an arbitrary score. The concept is shown in fig. 3. Just a small fluctuation around a stable mean does not necessarily give a warning signal. On the other hand, if an increase is persistent and is accompanied by abnormal environmental conditions, it represents a stronger basis to issue a warning.

C. Persistence Rule

The rule of persistence has been incorporated to safeguard against the influence of one abnormal prediction on the warning condition.

Let C_t be the number of consecutive instances meeting the escalation condition. The generic condition of alert may be expressed as

$$W_t = \text{ALERT} \quad \text{if} \quad P_t > \tau_p,\ \hat{\beta}_t > \tau_\beta,\ A_t > \tau_A,\ C_t \geq q. \quad (5)$$

The values of the parameters τ_p, τ_β, τ_A, and q have been calculated through validation. Thus, the application of the alert rule could be reproduced and it is not necessary to adjust manually the rule by the results of the final test series.

IX. Data Quality and Leakage Prevention

A. Timestamp Alignment

Various environmental datasets can be benchmarked against one another due to their different sampling frequencies. In order to allow comparisons, observations of different datasets are lined up at a common timestamp. When values need to be aggregated, only the information that could have been sampled previously is utilized. Timestamps of events are kept separately to allow for the creation of the target time window without introducing information from the future.

B. Missing Observations

There are missing values in satellite data, sensors, and landslide inventories. Short gaps can be filled using a verified time method, while longer gaps can be shown with indicators of missingness or omitted based on predetermined rules. This way, a missing-data experiment can show whether incomplete observations bring important alteration to warning progress.

TABLE V
Data Quality Controls

Issue	Control	Reason
Future information	Time-aware feature construction	Prevent leakage
Different sampling rates	Common temporal grid	Consistent alignment
Missing values	Validated imputation/indicator	Robustness
Spatial dependence	Spatial holdout	Generalization
Event duplication	Inventory deduplication	Correct targets
Scaling leakage	Training-set statistics	Fair evaluation

C. Spatial Leakage

Dividing neighboring observations at random may yield a mistaken impression of algorithm accuracy, as observations near each other tend to differ only slightly in terms of their environmental setting. If there is sufficient data to use, it is best to use test regions that are located far from each other. By employing such test regions one is able to check whether the models built on the basis of the training data collected can be generalized to different locations.

X. Operational Warning Workflow

A. Monitoring Cycle

In every monitoring cycle, the system picks up new observations and processes the feature vector to assess the current risk, develop the location-specific abnormality, and generate the risk trajectory. The warning state is then recorded with the relevant time of its measurement and the variables that contributed to it.

B. Warning Explanation

It must be ensured that the messages sent in the alerts are easily understood by people who do not have any technical knowledge. The dashboard must display the risk at the present moment, its change since the recent baseline, the total rainfall in a current time interval, how moist the soil is, and the risk trend. It may well be that some feature attribution methods would help in identifying the factors influencing the system predictions through the analysis of SHAP values. However, the explanations received must be regarded as evidence but not proof.

C. Alert Lifecycle

As an alerting system must ensure a warning process, it is important to clearly delineate between different processes occurring during an alerting event: starting the alert, keeping it active, reducing its level, and closing it. An alert may be kept active for as long as the escalation conditions still hold true. When the risk reduces for a specified time period, the

TABLE VI
SENSITIVITY STUDY FACTORS

Factor	Levels	Observed Effect
Risk threshold	Low/medium/high	Alert sensitivity
Trend threshold	Low/medium/high	Escalation sensitivity
Persistence	2/3/4+ steps	Noise suppression
Look-back window	Short/medium/long	Temporal context
Features	Full/reduced	Dependency analysis
Missingness	Low/medium/high	Robustness

system may return the alert to the ALERT mode and resume it later. Storing the history of alerts is important in terms of being able to analyze false alarms in the future easily.

XI. Comparative and Sensitivity Analysis

A. Baseline Comparison

The performance of the entire framework needs to be judged against a basic baseline which uses a prediction-only model. Baseline should, wherever possible, apply the original data set and prediction model without elements of time escalation and local abnormality detection. This makes measuring the contribution of the warning component straightforward.

B. Threshold Sensitivity

A single threshold cannot be universally established for different datasets and conditions of use. Sensitivity analysis allows one to change risk threshold and requirement for the duration of the alert and obtain the required compromise between successful warning and false alarms.

XII. Results and Analysis Plan

The performance of the numbers is conditional upon the ultimate geographical region of the research, the inventory of events, the forecasting period, and the composition of the tests, so the numeric data can be introduced only when the testing and analysis is over. The outcome of the analysis must contain the results associated with predictions, and the results aimed at warnings as well. In particular, the assessment of the risk trajectory must be carried out in order to verify the fact of warning issuance prior to events and discover environmental disturbances that cause excessive warnings.

XIII. Discussion

The key methodological feature of the suggested framework is that it considers risk to be a phenomenon that evolves over time. The interpretation at a particular point may be influenced by measurement error or some level of uncertainty in the model. By monitoring predictions over time and comparing situational conditions with locally-established baseline levels, the warning system can require supporting evidence before changing the level of the alert. This contributes to the interpretability of the decision-making process, reducing dependence on a fixed

TABLE VII
FINAL RESULTS TABLE TEMPLATE

Method	Prec.	Recall	F1	PR-AUC
Prediction only	–	–	–	–
Temporal layer	–	–	–	–
Complete system	–	–	–	–

TABLE VIII
WARNING-ORIENTED RESULTS TEMPLATE

Method	Detection	Lead Time	False Alarm
Prediction only	–	–	–
Temporal layer	–	–	–
Complete system	–	–	–

threshold. Location-adaptive baseline levels are particularly useful when the terrains and environments considered in the study differ. Specifically, something that may seem out of the ordinary in a particular location may be entirely normal somewhere else. However, the baseline creation process should be free from data leakage and should be managed carefully with time as long-term conditions change. This means that in reality baselines should be treated as an entity that is subject to validation, not as the history that works everywhere without questions.

Lead time should be looked at in conjunction with false alarms. Increasing the sensitivity of the system may produce earlier alerts, but it might also lead to a higher number of unnecessary alerts. A stricter rule may cut down on the false alarms but fail to identify fast-changing situations. Therefore, the evaluation proposed does not make use of a single measure, but reports a number of event-based measures.

XIV. Limitations and Future Work

The system relies on the precision and temporal resolution of environmental data and landslide catalogs. Evidence of events or their date of occurrence can greatly affect processes of model training and lead time estimation. Satellites can provide data which may have some gaps and uncertainty related to the measured values. Moreover, statistical forecasting of risks does not imply incorporation of causal relationships. Hence, the what-if test should be considered as the test of model vulnerability rather than deterministic physical modeling.

Future iterations of the system may involve ground deformation measurements, improved satellite imaging, physics-based constraints, spatial graph models, and chance forecasting. In addition, further research could analyze cross-area transfers,

concept drifts, adaptive calibration, and warning interfaces that match user requirements. Validation with disaster management partners at field level will ultimately be required before the tool can enter operation.

XV. Conclusion

In this paper, an AI-oriented and site-adaptive framework has been proposed for the early warning about landslides. With this approach, it is possible to combine the multi-temporal characteristics of the environment with machine-learning-based risk evaluation and abnormal-change detection. The proposed system is not treating every forecast as an independent indicator, but rather watches the risk and changes its forecasting state only when matching evidence reaches a certain level of reliability and duration. As for the proposed method for assessment, it consists of several parameters that allow judging about the predictive performance and lead time, missed events, uncertainties, the number of false alarms. In general, the new framework is created as a reproducible prototype for research and experiments where its overall numerical effectiveness would have been proven.

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